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Edge-Ai Enabled IOT Architecture for Early Fire Detection in Smart Agriculture

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ABSTRACT
Early detection of fire in agricultural environments is critical for preventing large-scale crop loss and ensuring safety. However, conventional fire detection systems are ineffective in open-field conditions due to delayed response, environmental interference, and high false alarm rates. This paper presents an Edge-AI enabled IoT architecture for real-time fire detection in smart agriculture.
The proposed system utilizes a Raspberry Pi as an edge computing platform to perform on-device processing, thereby reducing latency and eliminating reliance on cloud infrastructure. A YOLOv8-based deep learning model is employed for real-time visual fire detection, while environmental monitoring is achieved using a DHT11 temperature–humidity sensor and an infrared flame sensor. A multi-sensor fusion approach is implemented to enhance detection accuracy and reduce false positives. A lightweight Flask-based web dashboard provides real-time visualization of live video streams, sensor data, and alert notifications. Experimental results demonstrate that the system achieves high detection accuracy with low latency and reliable performance under various conditions. The proposed solution is cost-effective, scalable, and well-suited for deployment in rural and resource-constrained agricultural environments.

Keywords— Edge Computing, Internet of Things (IoT), Fire Detection, YOLOv8, Deep Learning, Smart Agriculture, Raspberry Pi, Computer Vision, Real-Time Monitoring, Sensor Fusion.

1. Introduction

Ensuring safety in agricultural environments is a critical challenge, particularly in regions prone to fire hazards. Agricultural fires can lead to extensive crop destruction, economic losses, and environmental damage. The rapid spread of fire in open-field conditions necessitates the development of efficient and reliable early detection systems.
Existing fire detection approaches are limited in their effectiveness when applied to agricultural settings. Conventional sensor-based systems, such as smoke and heat detectors, are primarily designed for indoor environments and perform poorly in outdoor conditions due to environmental variability [10]. Additionally, cloud-based fire detection solutions introduce latency and depend heavily on reliable communication infrastructure, which is often unavailable in rural areas [2]. These limitations highlight the need for intelligent, low-latency, and autonomous detection mechanisms.

Recent advancements in Edge Artificial Intelligence (Edge-AI), the Internet of Things (IoT), and deep learning have enabled the development of real-time monitoring systems. Deep learning-based object detection models, such as YOLO, have demonstrated high accuracy and efficiency in visual recognition tasks [5]. Furthermore, edge computing allows data to be processed locally on embedded devices, reducing latency, bandwidth usage, and dependency on cloud services [2].

In this work, an Edge-AI enabled IoT framework is proposed for early fire detection in smart agriculture. The system integrates deep learning-based visual detection with environmental sensing and performs local processing on an embedded platform. By combining multiple sensing modalities and edge intelligence, the proposed approach aims to deliver a scalable, cost-effective, and high-performance solution for real-world agricultural applications.

2. LITERATURE SURVEY

The development of effective fire detection systems has attracted significant research attention, particularly due to the growing need for intelligent monitoring in agricultural environments. Traditional fire detection methods primarily rely on smoke and heat sensors, which are designed for indoor applications. However, these systems are not well suited for open-field environments, as environmental factors such as wind, dust, and temperature variations can lead to delayed detection and reduced accuracy [10]. The integration of Internet of Things (IoT) technologies has enabled continuous monitoring and remote access to environmental data in agricultural systems. IoT-based fire detection solutions typically employ distributed sensors and cloud platforms for data processing and analysis. Despite these advantages, such systems often suffer from high latency and dependency on stable internet connectivity, limiting their effectiveness in rural and remote areas [2].

Recent advancements in artificial intelligence, particularly deep learning, have enabled the use of computer vision techniques for fire detection. Real-time object detection models such as YOLO have demonstrated high accuracy and efficiency in identifying fire from image and video data [5]. These approaches provide improved detection capabilities by enabling visual confirmation of fire events.

Furthermore, edge computing has enhanced the deployment of AI models on embedded devices, allowing local data processing and real-time decision-making. However, many existing solutions lack effective integration between environmental sensing and vision-based detection, which reduces overall system reliability. To address these limitations, this work proposes an Edge-AI enabled IoT system that combines multi-sensor data with real-time visual detection, providing a robust, low-latency, and efficient solution for fire detection in agricultural environments.

3. METHODOLOGY

The methodology of the proposed system focuses on integrating environmental sensing, computer vision, and edge computing to enable real-time fire detection in agricultural environments. The system follows a structured workflow consisting of data acquisition, processing, decision-making, and alert generation.

3.1 Data Acquisition

The system collects real-time data from multiple sources to ensure comprehensive monitoring. Environmental parameters such as temperature and humidity are measured using a DHT11 sensor, while an infrared-based flame sensor detects fire-related radiation. In parallel, a camera module continuously captures video frames from the monitored area. This multi-source data acquisition enhances the system's ability to accurately identify fire conditions.

3.2 Data Processing

All acquired data is processed locally on a Raspberry Pi, which functions as the edge computing unit. Video frames are analyzed using a YOLOv8 deep learning model to detect fire regions in real time. Meanwhile, sensor data is continuously updated and filtered to remove noise and improve reliability. Local processing significantly reduces latency and eliminates dependency on cloud-based systems.

3.3 Decision Mechanism

The system employs a hybrid decision-making strategy that combines visual detection with sensor-based inputs. A fire alert is triggered when the following conditions are met:

- The YOLO model detects fire with a confidence value above a predefined threshold
- The flame sensor detects infrared radiation

A high-confidence alert is generated when both conditions occur simultaneously. This fusion-based approach enhances detection accuracy while minimizing false positives.

3.4 Alert and Monitoring System

Upon detection of a fire event, the system generates real-time alerts through a Flask-based web dashboard. The dashboard provides live video streaming, sensor data visualization, and alert notifications. Visual indicators and event logs are updated instantly, enabling users to respond promptly and effectively.

4. HARDWARE IMPLEMENTATION

The hardware design of the proposed system focuses on the use of cost-effective and efficient components to support real-time fire detection in agricultural environments. The overall setup ensures reliability, ease of deployment, and suitability for large-scale field applications.

4.1 Processing Unit

A Raspberry Pi 4 is used as the central processing unit of the system. It performs multiple functions, including acquisition of sensor data, execution of the deep learning model for fire detection, and hosting of the web-based monitoring interface. Its compact size and sufficient computational capability make it well suited for edge-based applications.

4.2 Environmental Sensor (DHT11)

The DHT11 sensor is used to monitor temperature and humidity levels in the surrounding environment. These parameters are important indicators of fire-prone conditions. The sensor communicates with the Raspberry Pi through a digital interface, enabling periodic and reliable data acquisition.

4.3 Flame Detection Sensor

An infrared-based flame sensor is incorporated to detect fire through emitted radiation. It provides an early indication of fire presence and enhances system reliability when combined with vision-based detection. The sensor supports quick response to fire-related signals.

4.4 Vision Module

A camera module is deployed to capture continuous visual data from the monitored area. The captured frames are processed using the deep learning model to identify fire regions. This module enables accurate detection and localization of fire events in real time.

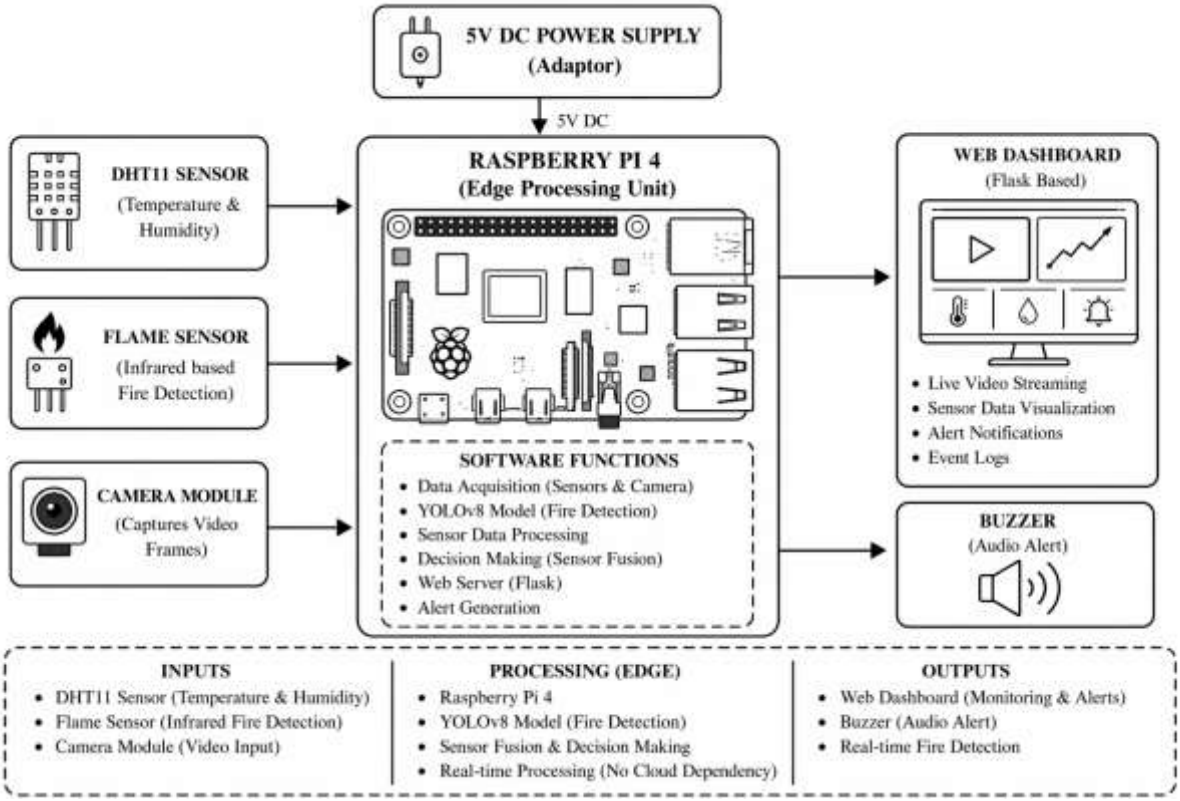
4.5 Power Unit

A regulated power supply is required to ensure stable operation of the Raspberry Pi and connected components. A 5V supply with adequate current capacity is used to maintain continuous and reliable system performance.

4.6 Interfacing and Connections

All hardware components are interfaced with the Raspberry Pi through GPIO pins and communication interfaces. The sensors are connected to designated input pins, while the camera module is interfaced through USB or a dedicated camera port. Proper wiring and grounding are ensured to maintain stable and error-free operation.

Figure 1: Hardware Architecture of the Proposed Edge-AI Fire Detection System



5. SOFTWARE IMPLEMENTATION

The proposed system employs a modular software architecture to enable efficient and real-time fire detection in agricultural environments. The implementation is developed using Python, which offers extensive support for computer vision, machine learning, and IoT-based applications.

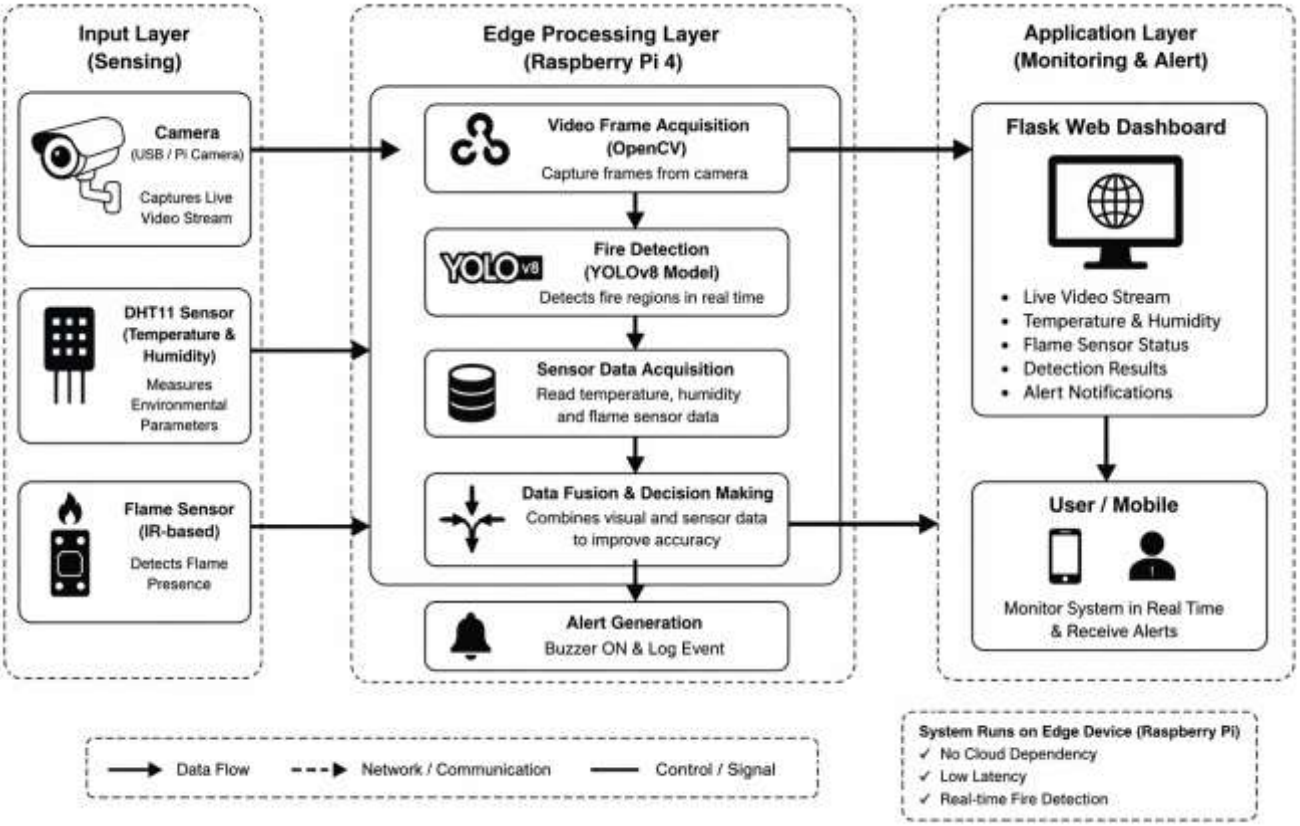
The system integrates multiple functional modules into a unified processing pipeline. Video data captured from the camera is processed using a YOLOv8-based deep learning model to detect fire regions in real time. Simultaneously, environmental data such as temperature, humidity, and flame sensor readings are continuously acquired and updated. The integration of visual and sensor data enhances the overall reliability and accuracy of the detection process.

To ensure smooth and uninterrupted operation, the system adopts a multi-threaded architecture. Separate threads are dedicated to video processing, sensor data acquisition, and web server management, enabling concurrent execution without performance degradation.

A web-based interface is developed using the Flask framework to facilitate user interaction and system monitoring. The dashboard provides real-time visualization of live video streams, sensor data, and alert notifications. The system operates entirely on an edge device, eliminating dependency on cloud services and significantly reducing latency.

To achieve efficient real-time performance on a resource-constrained edge device, several optimization techniques are employed. The input video resolution is adjusted to balance detection accuracy and processing speed. The YOLOv8 model is configured to operate in a lightweight mode suitable for embedded deployment. Additionally, frame skipping and optimized data handling techniques are used to reduce computational overhead. These optimizations ensure that the system maintains low latency while delivering reliable fire detection performance. Overall, the software implementation ensures efficient execution, scalability, and reliable performance, making it well suited for real-time fire detection applications in smart agriculture.

Figure 2: Overall Software Architecture and Workflow of the Proposed Edge-AI Fire Detection System



6. RESULTS AND DISCUSSION

6.1 Experimental Setup

The proposed Edge-AI fire detection system was evaluated under controlled outdoor conditions to simulate real agricultural environments. The system was deployed using a Raspberry Pi 4 integrated with a camera module, DHT11 sensor, and flame sensor. Various fire scenarios were created, including small flame sources, burning agricultural materials, and non-fire conditions such as bright sunlight and heated objects. Testing was conducted at different distances ranging from 0.5 m to 3 m to analyze detection capability Performance metrics such as detection latency, detection accuracy, false positive rate, and system resource utilization were recorded during the experiments.

6.2 Detection Performance

The system demonstrated reliable performance in detecting fire events at close and medium distances. Fire

detection was achieved within 1 second at short distances (0.5 m), while detection time increased slightly with distance. The YOLOv8 model achieved a high detection accuracy of approximately 92% for distances up to 2 meters. Beyond this range, detection confidence gradually decreased due to reduced visibility of flame features. False positive occurrences were minimal, with fewer than two incorrect detections per hour under challenging lighting conditions, indicating robustness of the model.

6.3 System Performance Analysis

The system maintained stable operation under continuous execution. The YOLOv8 model achieved an inference speed of approximately 4–6 frames per second on the Raspberry Pi. CPU utilization remained high (85–95%) during active detection due to continuous image processing, while memory usage stayed within acceptable limits (1.2–1.8 GB). The dashboard response time was observed to be less than 1.5 seconds, ensuring near real-time alert visualization. The system successfully operated continuously for 48 hours without failure, demonstrating reliability.

TABLE 1: Performance Evaluation of Proposed System

Metric	Observed Value
Detection Latency (0.5 m)	< 1 second
Detection Latency (2 m)	1 – 3 seconds
Detection Latency (3 m)	2 – 5 seconds
True Detection Rate	~92% (up to 2 m)
False Positive Rate	< 2 per hour
YOLO Inference Speed	4 – 6 FPS
Dashboard Latency	< 1.5 seconds
System Uptime	48 hours continuous

TABLE 2: System Resource Utilization

Parameter	Value
CPU Usage	85 – 95%
RAM Usage	1.2 – 1.8 GB
Processing Device	Raspberry Pi 4
Model Used	YOLOv8n

6.4 Sensor and Environmental Analysis

Environmental sensor data provided valuable support for fire detection. Temperature readings showed a gradual increase prior to fire detection, often rising 30–60 seconds before visual confirmation. Humidity levels exhibited a decreasing trend during fire events, particularly in dry conditions. These patterns highlight the importance of integrating environmental sensing with vision-based detection. The flame sensor proved effective in detecting fire during low-light and nighttime conditions, compensating for reduced visibility in camera-based detection.

Figure 3: Temperature and Humidity Variation During Fire Event

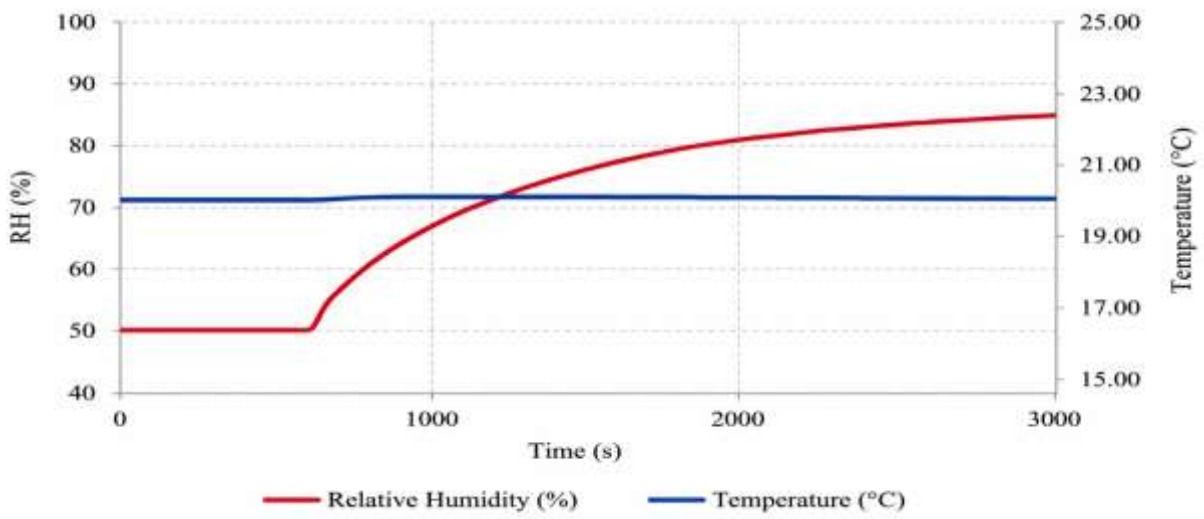
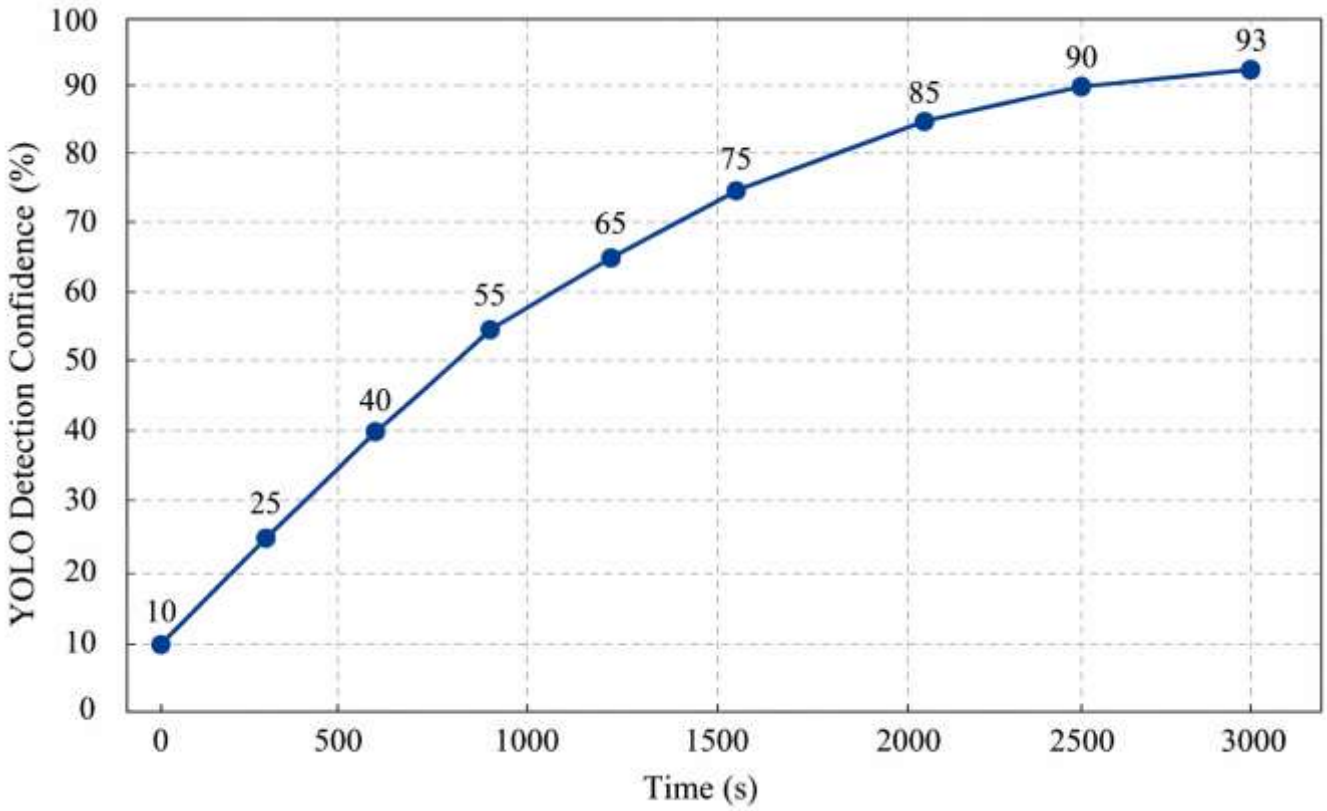


Figure 4: YOLO Fire Detection Confidence Over Time



6.5 Discussion

The experimental results confirm that the integration of Edge-AI with multi-sensor data significantly enhances fire detection performance. The combination of YOLO-based visual detection and sensor fusion reduces false alarms and improves reliability compared to traditional systems. Edge processing eliminates dependency on cloud infrastructure, resulting in reduced latency and improved system responsiveness. This makes the system highly suitable for rural agricultural environments where internet connectivity is

limited. Overall, the proposed system provides a cost-effective, scalable, and efficient solution for real-time fire detection in smart agriculture.

7. CONCLUSION

This paper presents an Edge-AI based fire detection system designed for real-time monitoring in agricultural environments. The proposed approach integrates a YOLOv8-based deep learning model with environmental sensors to enable accurate and reliable fire detection. By performing all computations on a Raspberry Pi-based edge device, the system achieves low latency and operates independently of cloud infrastructure. Experimental results demonstrate that the system achieves high detection accuracy with reduced false positives, owing to the fusion of visual and sensor data. The implementation on the Raspberry Pi platform further confirms the feasibility of deploying advanced AI models in resource-constrained environments. However, the system's performance can be further improved for long-range detection and varying environmental conditions. In summary, the proposed solution provides a scalable, cost-effective, and efficient framework for early fire detection, contributing to enhanced safety and smart agricultural management. Furthermore, the system highlights the practical applicability of integrating Edge-AI and IoT technologies using low-cost embedded platforms such as Raspberry Pi. The modular design enables easy scalability and deployment across large agricultural areas. By combining real-time monitoring with intelligent decision-making, the system supports proactive risk management and rapid response. This work emphasizes the potential of edge-based intelligent systems in improving reliability, reducing response time, and enabling sustainable smart agriculture practices.

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